Thread Pools and Concurrency Problems

Thread Pools and Concurrency Bugs

Learning Objectives:

- Be able to describe the structure of a thread pool and why we use them
- Be able to identify concurrency bugs by thinking concurrently
- Be able to identify classes of concurrency bugs given their description

Thread Pool

- System creates a number of worker threads.
- Each worker thread repeatedly:
  - Pops items off of a shared queue
  - Processes the item
- A producer thread (often main):
  - Repeatedly pushes items onto the shared queue
- Advantages compared to fork-join parallelism:
  - Less expensive since fewer calls to creating fork.
  - Support for “bursts” -> The queue allows you to have a backlog.
- Demo: See code in class!

Thinking Concurrently

- Bugs can crop up when threads operate concurrently!
- High level approach to finding them:
  - Identify all shared states
  - Look for compound operations (e.g., Read, modify, write operations)
  - Look for instances where more than one thread writes or reads to the same variable.
  - Manually simulate different schedules. *Think: what could an adversarial scheduler do?*
- Example: